

Design Procedure of Subbandgap Voltage Reference for Energy Harvesting Power Management Unit in 65 nm CMOS Technology

Sonia Salhi^a, * and Sid Ahmed Tedjini^a

^a Division Microélectronique et Nanotechnologie, Centre de Développement des Technologies Avancées, Algiers, Algeria

*e-mail: ssalhi@cdta.dz

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Abstract—This paper presents the design procedure of subbandgap voltage reference intended for energy harvesting systems. The circuit provides a stable nominal output voltage of 1 V, cross supply voltage range from 1.4 to 2.5 V. The best achieved temperature coefficient is 35 ppm/°C, with an average value of 48 ppm/°C over 12 samples V_{dd} . The designed circuit exhibits strong immunity against supply voltage fluctuation and good overall stability, with an average power supply rejection ratio of 54 dB. Implemented in 65 nm CMOS technology, the circuit occupies an area of $280 \times 213 \mu\text{m}^2$ and consumes an average current of 32 μA under supply voltage range from 1.4 to 2.5 V.

Keywords: bandgap voltage reference, PTAT, CTAT

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1. INTRODUCTION

The growing ubiquity of smart devices in modern life, underpinned by advancements in miniaturization and wireless connectivity, and the pursuit of smarter lifestyles, has led to a surge in demand for applications such as the Internet of Things (IoT), wireless sensor nodes, wearable and implantable devices. These technologies, often constrained by the need for reliable low-voltage operation, rely on architectures powered by batteries or energy-harvesting. Most such architectures incorporate bandgap voltage reference circuit designed to generate certain stable voltage that serves in various functional blocks such as power management unit, RF and communication module, LDO, ADC, DAC, clock reference, and other precision analog circuits [12].

In the power management unit showed in Fig. 1, the BGR has an impact on the overall performance of the system, that must provides a stable voltage with minimal dependence on temperature, supply voltage,

and process variation to the LDO, ensuring a good system functionality [1, 5].

In the literature, the BGR is most often used as a voltage reference due to its high accuracy and stability against Zener-diode-based reference, buried Zener reference, and XFET [4, 13, 19]. The first successful bandgap voltage reference was introduced by Wildar, exploiting the precise opposition between two temperature dependent phenomena, the negative temperature coefficient of the base-emitter voltage (V_{BE}) of a bipolar transistor and the positive temperature coefficient of base-emitter differential voltages of two transistors biased at different current densities. The combination of these complementary characteristics allowed to generate a voltage reference with little sensitivity to temperature variations [3, 6, 17, 18].

Modern low-power electronics operate at supply voltage of 1.2 V or below. However, conventional bandgap reference is fundamentally constrained by 1.2 V limit imposed by the intrinsic bandgap energy of silicon. Consequently, the design of subbandgap voltage

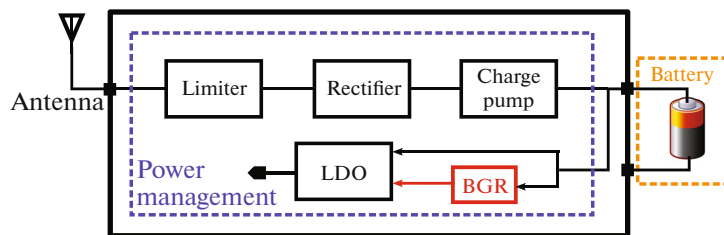


Fig. 1. Power management unit in energy harvesting system.

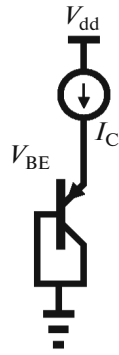


Fig. 2. CTAT voltage generation circuit.

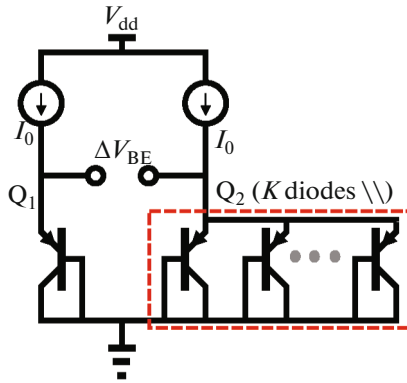


Fig. 3. PTAT voltage generation circuit.

with a good trade-offs between temperature, supply voltage, and process variations presents a critical design challenge. In this context, this paper aims to describe design methodology for subbandgap voltage reference (<1.2 V) implemented in a submicron CMOS technology process, specifically intended for integration within low-dropout regulators (LDO) in power management systems [12, 14, 18].

The paper is organized as follows, Section II, covers the fundamental principles of Bandgap voltage reference, Section III details the proposed design procedure, followed by the post-layout simulation results and discussion in Section IV, and the last section provides the conclusion.

2. FUNDAMENTAL PRINCIPLE OF BANDGAP REFERENCES

The bandgap reference circuit achieves a nominally constant voltage independent of process, supply voltage and temperature variations. Its core mechanism combines two circuits exhibiting complementary temperature coefficients to cancel temperature dependence. Let us define each circuit.

2.1. Complementary to Absolute Temperature (CTAT)

Figure 2 presents a negative temperature coefficient voltage generation circuit based on a bipolar transistor connected diode biased with a constant current (I_C). The voltage across base-emitter (V_{BE}) is expressed as follows,

$$V_{BE} = nV_T \ln\left(\frac{I_C}{I_s}\right), \quad (1)$$

where, n is the emission coefficient, V_T is a thermal voltage, and I_s is emitter saturation current given by the following expression,

$$I_s = bT^{4+m} \exp\left(\frac{-E_g}{kT}\right), \quad (2)$$

where, b is a proportionality factor, $m \approx 3/2$, k is the Boltzmann constant, T temperature, and E_g the bandgap energy of silicon.

Based on the expressions (1) and (2), V_{BE} depends on bandgap energy of silicon and temperature. Thus, the variation of the base-emitter voltage with temperature is expressed as follows.

$$\frac{\partial V_{BE}}{\partial T} = \frac{V_{BE} - (4+m)V_T - \frac{E_g}{q}}{T}, \quad (3)$$

where, q is an electron charge.

The expression (3) reveals that the base-emitter voltage of the BJT transistor varies inversely with temperature.

2.2. Proportional to Absolute Temperature (PTAT)

Considering two BJT transistors connected as diode, (Q_1 , and Q_2), operating under the same current as depicted in Fig. 3. The transistor Q_2 is designed to be larger than Q_1 , with an emitter area ratio K . Consequently, Q_2 exhibits a smaller voltage drop than Q_1 , for the same current through each branch, as a result, the base-emitter voltage difference between the two BJT ΔV_{BE} is given in (4).

$$\Delta V_{BE} = V_T \ln(K). \quad (4)$$

From expression (4), the ΔV_{BE} is proportional to the thermal voltage, which increases with temperature.

Thus, its voltage variation with temperature is given by (5).

$$\frac{\partial \Delta V_{BE}}{\partial T} = \frac{k}{q} \ln(K). \quad (5)$$

The expression (5) reveals that the base-emitter voltage difference variation is proportional to temperature.

3. SUBBANDGAP REFERENCE DESIGN METHODOLOGY

The subbandgap circuit generates an output voltage below silicon's bandgap voltage (1.2 V) through a

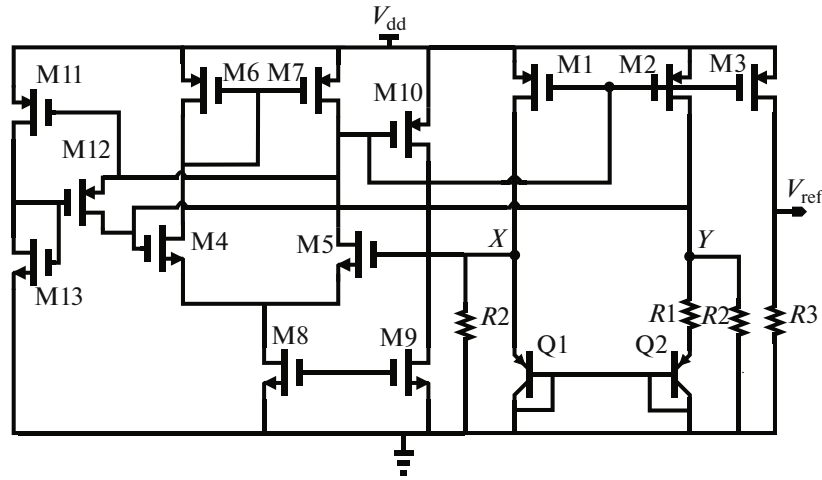


Fig. 4. Bandgap voltage reference circuit.

combination PTAT, CTAT circuits, op amp, and resistive elements. The resistors R_1 , and R_2 are introduced to achieve temperature compensation, while resistor R_3 regulates the output voltage. as depicted in the Fig. 4.

To mitigate the power supply fluctuation, a high gain amplifier with wide input common-mode voltage range is employed. Where the op amp's output connects to the gates of M1 and M2 to control current through branch of Q1 and R_1 with Q2, ensuring equal voltage at the nodes X and Y. However, the start-up circuit (M11, M12, M13) is required to avoid scenario when M1, M2 gate voltage is equal to V_{dd} resulting in zero current.

Circuit analysis reveals that the current flowing through R_1 , determined by the ΔV_{BE} , constitutes a PTAT function expressed in (6). Conversely, the current through R_2 exhibits CTAT characteristics as given in (7). As a results, the total current becomes temperature independent.

$$I_{PTAT} = \frac{nV_T \ln(K)}{R_1}, \quad (6)$$

$$I_{CTAT} = \frac{V_{BE1}}{R_2}. \quad (7)$$

The output voltage reference is generated by voltage across the resistor R_3 . Where transistor M3 mirrors the combined PTAT and CTAT currents given by (6) and (7), respectively, which flows trough R_3 . The expression of the output voltage reference is given by the following expression,

$$V_{ref} = \frac{R_3}{R_1} \left(nV_T \ln(K) + \frac{R_1}{R_2} V_{BE1} \right). \quad (8)$$

The temperature behavior of the voltage reference is derived from (8) and given by the following expression,

$$\frac{\partial V_{ref}}{\partial T} = \frac{R_3}{R_1} n \ln(K) \frac{\partial V_T}{\partial T} + \frac{R_3}{R_2} \frac{\partial V_{BE1}}{\partial T}. \quad (9)$$

The temperature variation in expression (9) depends on thermal voltage, and base-emitter voltage. The resistance ratio is the key critical parameter for temperature compensation, enabling mutual compensation between thermal voltage and base-emitter voltage.

3.1. Application for Energy Harvesting Power Management Unit

The BGR voltage circuit is designed for integration into the power management system of passive smart card. In this system, the power is transferred from the reader to the card's antenna based on the inductive decoupling technique. The supply voltage derived from the tag antenna varies with inductive strength and the reader tag distance. Consequently, the system is supplied by a RF harvester charged in storage capacitor which is variable with the harvested and consumed power. The BGR receive the supply voltage from storage capacitor and must be generate PVT insensitive reference voltage to the LDO's error amplifier to ensure stable 1.2 V DC output to supply the whole of system. Therefore, the designed BGR must provide voltage reference lower than the LDO's output voltage. In this context, the 1 V is the target reference voltage, where the design sizing parameters are computed thanks to the design procedure defined from expressions (6) to (8).

The Table 1 summarizes a computed devices parameters, for a current mirror designed to generate 1.5 μA bias current. Furthermore, to ensure robust power sup-ply rejection ratio performance, the Op Amp is designed with a minimum gain of 60 dB across an operational voltage range.

4. SIMULATION AND RESULTS

Following the defined design procedure outlined above, the BGR circuit depicted in Fig. 4 is imple-

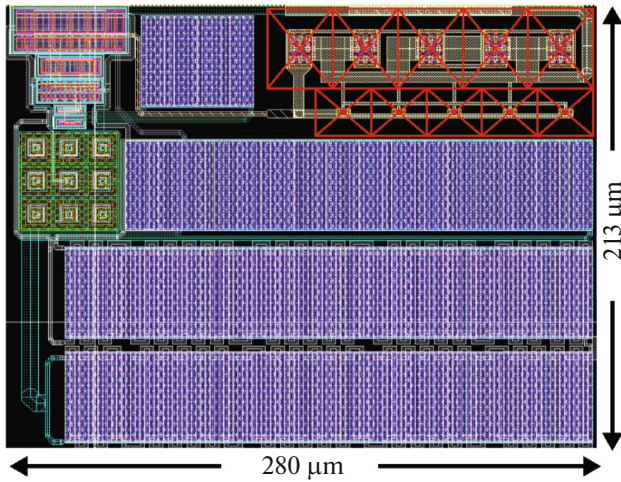


Fig. 5. BGR layout implemented in 65 nm CMOS technology.

mented in 65 nm CMOS process and simulated using SPECTRE simulator. The thick oxide MOS transistors are used to avoid oxide damage caused by high supply voltage (storage energy). The circuit operates with supply voltage ranging from 1.4 to 2.5 V, with temperature range of 0 to 100°C. The layout photography of the implemented BGR circuit shows in Fig. 5. The circuit consumes $280 \times 213 \mu\text{m}^2$ of the silicon active area with average current dissipation of 32 μA .

4.1. Output Voltage Variation vs. Temperature and V_{dd}

The post layout simulation results for output voltage reference variation with the temperature and V_{dd} are interpreted in Fig. 6.

Figure 6a reveals the voltage reference turns on at V_{dd} of approximately 1.4 V. The Close-up behavior of voltage reference is given in Fig. 6b. Based on these results and temperature behavior depicted in Fig. 6c, the temperature coefficient (TC) is derived. The best case is approximately 36 ppm/°C that is reached when the storage capacitance is fully charged at 2.5 V. As the storage capacitance discharge due to power consumption, the TC gradually increases, reaching its worst case value of 63 ppm/°C at 1.4 V, with average TC of 48 ppm/°C.

4.2. Stability and Output Voltage Ripple

The susceptibility of the output reference voltage to supply voltage fluctuations is quantified by the power supply rejection ratio (PSRR), a key figure of merit for supply suppress fluctuation. Post-layout simulation results, presented in Fig. 7, show PSRR value ranging from -40 to -60 dB at 100 Hz across the operating range, this indicates effective mitigation of the supply ripple.

Complementing the PSRR analysis, a system stability is evaluated through gain and phase margins

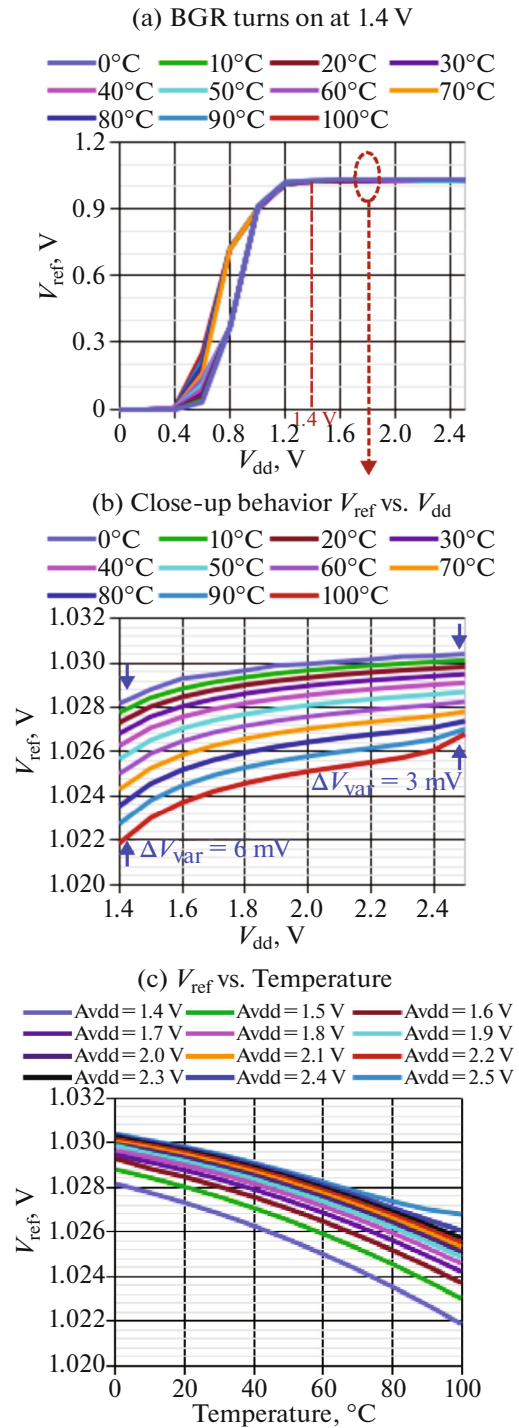


Fig. 6. V_{ref} behavior temperature and V_{dd} .

Table 1. Devices parameters for 1 V target voltage reference

Device	Computed parameters value
K	8
R_1	35 k Ω
R_2	319 k Ω
R_3	263 k Ω

summarized in Table 2. Across the operating range, the gain margin remains positive at approximately 24 dB while the phase margin maintains 74° , ensuring stability criteria. In addition, the op amp exhibits 103 dB average DC gain, ensuring matched current through branches X and Y, at a result, a good stability and robust rejection to the power supply fluctuation.

Table 2 extends the circuit performance beyond to gain margin, phase margin, and op amp DC gain to include current dissipation across the V_{dd} range (1.4–2.5 V) for 12 samples. The postlayout simulation reveals an average current dissipation about $32 \mu\text{A}$.

4.3. Output Noise

Figure 8 presents the output noise spectral density postlayout simulation of the BGR. The result reveals a noise spectral density of $5 \mu\text{V}/\sqrt{\text{Hz}}$ @ 100 Hz.

4.4. Process Corners

Figure 9 depicts BGR's process variation of postlayout simulation across FF, SS, FS, and SF corners over the full V_{dd} and temperature ranges. The output voltage reference maintains a stable nominal voltage of 1 V with minimal variation under all conditions. The worst case voltage variation is 7 mV occurred in the SF corner at minimum V_{dd} (low battery), while the best case voltage variation is 2.3 mV observed in the FF corner at maximum V_{dd} (full battery). Consequently, the voltage reference exhibit monotonic behavior across PVT corners.

4.5. Performance vs. Prior Bandgap Based Designs

Table 3 summarizes the performance of the proposed voltage reference circuit in comparison with against prior bandgap based designs. The circuit achieves an average temperature coefficient of $48 \text{ ppm}/^\circ\text{C}$, which is on par with [16] and [8]. It offers an average PSRR of 54 dB, providing better and more stable supply rejection than [16], [2], and [7], though slightly below the performance reported in [11], [10], and [9], but remains competitive with [13]. The average current consumption of $32 \mu\text{A}$ higher than ultralow power designs in [16] and [8], but remains competitive with [10], and [9], and significantly better than [7]. Overall, this design achieves optimal trade-offs between temperature variation, supply voltage fluctuation, and power dissipation while maintaining robust stability according to technology scaling effects, positioning it as viable solution for moderate precision systems such as energy harvesting applications.

CONCLUSIONS

This work presented a design procedure and implementation for 1 V subbandgap voltage reference in 65 nm CMOS technology. The postlayout simula-

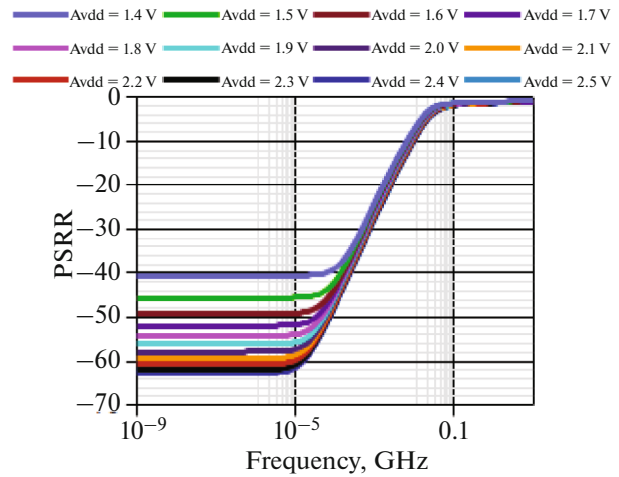


Fig. 7. Power supply rejection ratio.

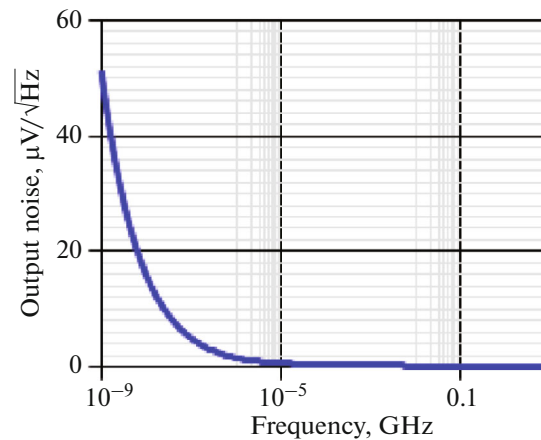


Fig. 8. BGR output noise spectral density.

Table 2. Bandgap voltage reference performance

V_{dd} , V	PSRR, dB	GM, dB	PM, deg	Op Amp gain, dB	I Supply, μA
1.4	-40.0	21.30	74.42	87.49	-25.86
1.5	-45.0	21.37	74.45	90.8	-26.67
1.6	-48.6	21.46	74.49	93.98	-27.63
1.7	-51.3	21.54	74.51	97.12	-28.70
1.8	-53.5	21.60	74.49	100.3	-29.87
1.9	-55.3	21.65	74.42	103.6	-31.12
2	-57.0	21.69	74.35	107.2	-32.44
2.1	-58.4	21.72	74.29	111.6	-33.82
2.2	-59.7	21.75	74.25	119.5	-35.26
2.3	-60.8	21.77	74.21	122.1	-36.74
2.4	-61.7	21.79	74.19	108.4	-38.26
2.5	-62.0	21.81	74.19	98.57	-39.83

GM: gain margin, PM: phase margin.

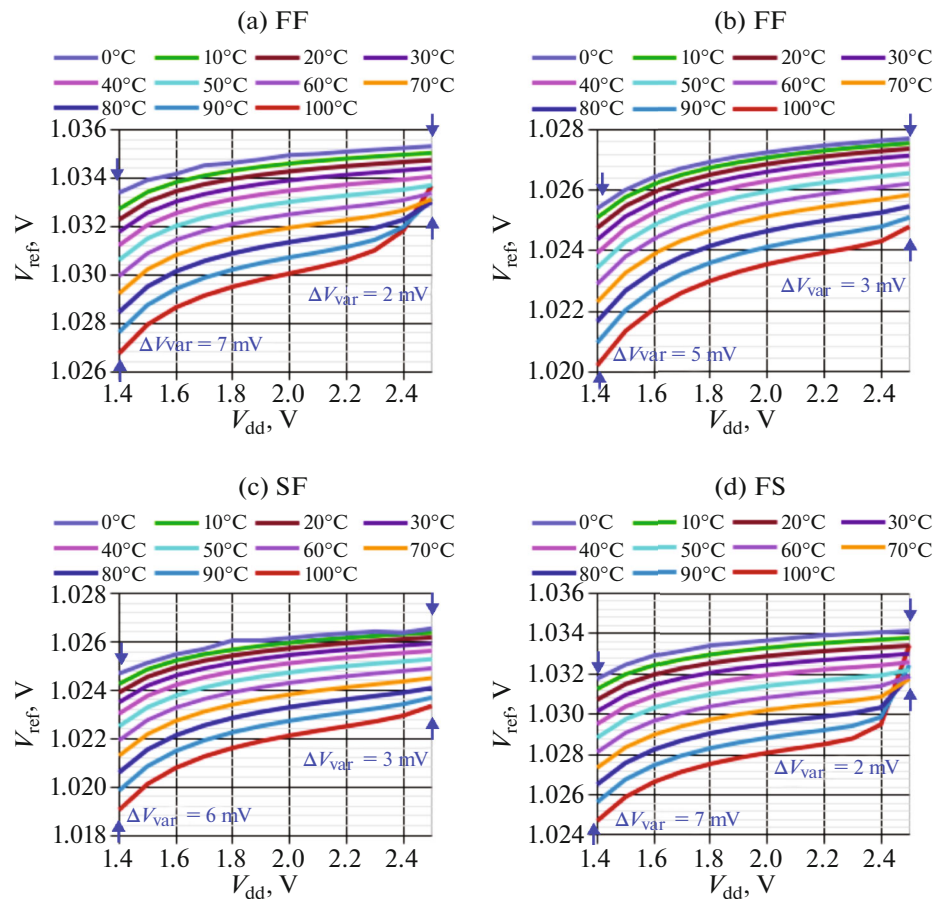


Fig. 9. Process variation (corner).

tion results reveal minimal output voltage variation against temperature, supply voltage and process corners, achieving an average temperature coefficient of 48 ppm/°C, 56 dB PSRR and an average current

consumption of 32 μ A across a supply voltage range of 1.4 to 2.5 V and a temperature range of 0 to 100°C. These performances are competitive and well-suitable for energy harvesting applications.

Table 3. Bandgap voltage compared with prior design

REF	Tech CMOS, nm	V_{dd} range, V	Temp range, °C	V_{ref} , V	PSRR, dB	TC, ppm/°C	Power or I consumption		Active area, mm ²
[16]	65	0.75 to 1.2	−40 to 90	0.474	−40	40	0.29	μ W	0.0198
[2]	180	0.4 to 1.8	−40 to 130	0.212	−40	84.5	—		0.09
[11]	40	1.2	−40 to 90	0.8	−71	12.51	9.6	μ W	—
[8]	90	1.2	0 to 100	0.72	—	53.1	0.58	μ W	0.027
[10]	500	2.1 to 5	−5 to 125	1.196	−84	9.72	38	μ A	0.053
[9]	180	3 to 5	−45 to 125	1.25	−100	4.2	127.8	μ W	1.03
[7]	65	1.04 to 1.2	—	0.571	−34.94	12.34	182.3	μ A	—
[15]	350	2.5 to 5	5 to 100	1.23	−57	8.5	—		0.034
This work	65	1.4 to 2.5	0 to 100	1	−54.4	48	32	μ A	0.059

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

REFERENCES

1. Ali Akbar, M., Zgaren, M., Ait, A., Ali, S., Amira, A., Benammar, M., Bensaali, F., Sawan, M., and Bermak, A., Gas identification using passive uhf rfid sensor platform, *Sensors and Transducers*, 2015, vol. 194, pp. 42–53.
2. Basyurt, P.B., Bonizzoni, E., Aksin, D.Yi., and Maloberti, F., A 0.4-V supply curvature-corrected reference generator with 84.5-ppm/°C average temperature coefficient within –40°C to 130°C, *IEEE Trans. Circuits Syst. II: Express Briefs*, 2017, vol. 64, no. 4, pp. 362–366. <https://doi.org/10.1109/tcsii.2016.2562738>
3. Brokaw, A.P., A simple three-terminal IC bandgap reference, *IEEE J. Solid-State Circuits*, 1974, vol. 9, no. 6, pp. 388–393. <https://doi.org/10.1109/JSSC.1974.1050532>
4. Bucur, V., Banarie, G., Marinca, S., and Bodea, M., A Zener-based voltage reference design compensated using a ΔV_{BE} stack, *2018 25th International Conference Mixed Design of Integrated Circuits and System (MIXDES)*, Gdynia, Poland, 2018, IEEE, 2018, pp. 116–120. <https://doi.org/10.23919/MIXDES.2018.8436687>
5. Han, P., Zhang, Zh., Xia, Ya., and Mei, N., A 920-MHz dual-mode receiver with energy harvesting for UHF RFID tag and IoT, *Electronics*, 2020, vol. 9, no. 6, p. 1042. <https://doi.org/10.3390/electronics9061042>
6. Hilbiber, D., A new semiconductor voltage standard, *1964 IEEE International Solid-State Circuits Conference. Digest of Technical Papers*, Philadelphia, 1964, IEEE, 1964, vol. 7, pp. 32–33. <https://doi.org/10.1109/isscc.1964.1157541>
7. Jain, Sh., Kanchetla, V.K., and Zele, R., A sub-1V, current-mode bandgap voltage reference in standard 65 nm CMOS process, *2022 IEEE 15th Dallas Circuit And System Conference (DCAS)*, Dallas, TX, 2022, IEEE, 2022, pp. 1–5. <https://doi.org/10.1109/dcass3974.2022.9845585>
8. Lee, K.K., Lande, T.S., and Häfliger, P.D., A sub- μ W bandgap reference circuit with an inherent curvature-compensation property, *IEEE Trans. Circuits Syst. I: Regular Pap.*, 2015, vol. 62, no. 1, pp. 1–9. <https://doi.org/10.1109/TCSI.2014.2340553>
9. Li, X., Li, J., Qiao, M., and Zhang, B., Curvature-compensated bandgap voltage reference with low temperature coefficient, *Electronics*, 2024, vol. 13, no. 22, p. 4490. <https://doi.org/10.3390/electronics13224490>
10. Ming, X., Hu, L., Xin, Y.-L., Zhang, X., Gao, D., and Zhang, B., A high-precision resistor-less CMOS compensated bandgap reference based on successive voltage-step compensation, *IEEE Trans. Circuits Syst. I: Regular Pap.*, 2018, vol. 65, no. 12, pp. 4086–4096. <https://doi.org/10.1109/tcsi.2018.2834468>
11. Mu, Sh. and Chan, P.K., Design of precision-aware subthreshold-based MOSFET voltage reference, *Sensors*, 2022, vol. 22, no. 23, p. 9466. <https://doi.org/10.3390/s22239466>
12. Razavi, B., *Design of Analog CMOS Integrated Circuits*, Electrical Engineering Series, McGraw-Hill, 2001.
13. Nasraty, R., XFET™ references low noise, lower voltage than Zeners, Micropower, better than bandgaps, *Analog Devices*, 1998, vol. 32, no. 3, pp. 1–4.
14. Salhi, S., Tedjini, S.A., and Slimane, A., 1v, 12ppm/°C bandgap reference circuit design in cmos technology, *2025 3rd International Conference on Electronics, Energy and Measurement (IC2EM)*, Algiers, 2025 (IEEE, 2025), pp. 1–5. <https://doi.org/10.1109/IC2EM63689.2025.11101163>
15. Shih, M.-H., Wu, H.-H., and Wei, C.-L., Design of wide-supply-voltage-range bandgap reference circuits for voltage regulators, *2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017–ECCE Asia)*, Kaohsiung, Taiwan, 2017, IEEE, 2017, pp. 1867–1871. <https://doi.org/10.1109/ifeec.2017.7992333>
16. Tan, X.L., Chan, P.K., and Dasgupta, U., A sub-1-V 65-nm MOS threshold monitoring-based voltage reference, *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, 2015, vol. 23, no. 10, pp. 2317–2321. <https://doi.org/10.1109/TVLSI.2014.2361766>
17. Widlar, R., Some circuit design techniques for linear integrated circuits, *IEEE Trans. Circuit Theory*, 1965, vol. 12, no. 4, pp. 586–590. <https://doi.org/10.1109/tct.1965.1082512>
18. Widlar, R.J., New developments in IC voltage regulators, *IEEE J. Solid-State Circuits*, 1971, vol. 6, no. 1, pp. 2–7. <https://doi.org/10.1109/jssc.1971.1050151>
19. Witt, T.J., Reymann, D., and Avrons, D., The stability of some Zener-diode-based voltage standards, *IEEE Trans. Instrum. Meas.*, 1995, vol. 44, no. 2, pp. 226–229. <https://doi.org/10.1109/19.377817>

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